

Introduction to Servlets in Java

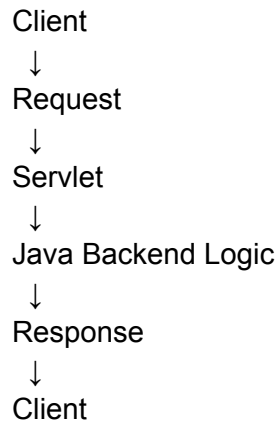
1. What is a Servlet?

A **Servlet** is a Java program that runs on a server.

It is used to:

1. Receive requests from the client.
2. Process the request using Java backend code.
3. Send a response back to the client.

Simple example



For example, when you open:

`http://localhost:8080/hello`

your browser sends a request to the server. The Servlet receives the request, processes it, and sends a response such as:

Hello, Manish!

2. Client and Server

Client

A **client** is the device or application that sends a request to a server.

Examples:

- Google Chrome
- Firefox
- Mobile apps
- Postman

For example, when you enter a URL in Chrome, Chrome acts as the **client**.

Server

A **server** is a computer or software that receives requests from clients and processes them.

For example:

Browser → Request → Server
Browser ← Response ← Server

3. Types of Servers

There are mainly two types we should understand here.

A. Web Server

A **web server** handles web-related requests.

Examples:

- Apache Tomcat
- GlassFish
- Jetty

For learning Servlets, we commonly use **Apache Tomcat**.

B. Application Server

An **application server** provides more advanced features for large applications.

Examples:

- JBoss
- WebLogic
- IBM WebSphere

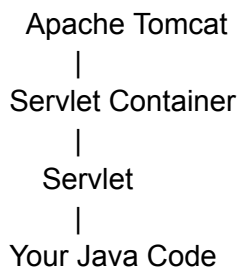
For now, you don't need to study application servers deeply.

4. What is Apache Tomcat?

Apache Tomcat is a server that can run Java web applications and Servlets.

It provides a **Servlet Container** that manages Servlets.

Think of Tomcat like this:



So, when we create a Servlet, **Tomcat runs and manages it for us.**

5. What is a Servlet Container?

A **Servlet Container** is a part of the server that manages Servlets.

Its main responsibilities include:

- Creating Servlet objects
- Managing the Servlet lifecycle
- Receiving HTTP requests
- Sending requests to the correct Servlet
- Providing request and response objects
- Sending the response back to the client
- Destroying the Servlet when it is no longer needed

In simple words:

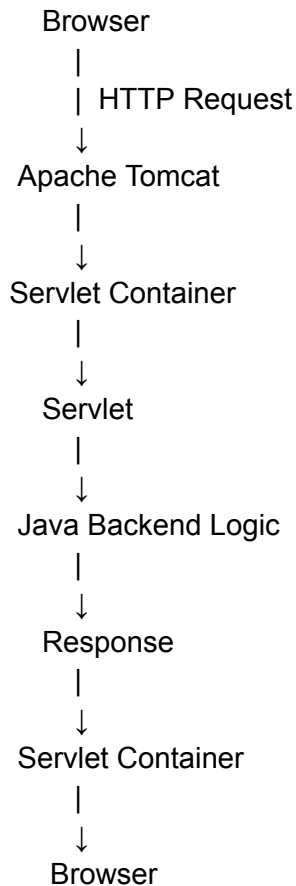
Servlet Container is responsible for running and managing Servlets.

6. How Does a Servlet Work?

Suppose you open:

`http://localhost:8080/hello`

The process looks like this:



Step-by-step

Step 1: The client sends a request.

Browser → /hello

Step 2: Tomcat receives the request.

Step 3: The Servlet Container checks which Servlet should handle `/hello`.

Step 4: The correct Servlet is executed.

Step 5: The Servlet processes the request using Java code.

Step 6: The Servlet creates a response.

Step 7: Tomcat sends the response back to the browser.

7. Can a Servlet Generate HTML?

Yes.

A Servlet can generate a dynamic response.

For example:

```
Hello Manish  
Welcome to my website  
Your account balance is ₹5000
```

The response can also contain HTML.

For example:

```
<h1>Welcome Manish</h1>
```

This is called **dynamic content** because the Servlet can generate different responses depending on the request or data.

8. Other Server-Side Technologies

Servlets are not the only technology used for backend development.

Java-based technologies

- Servlet
- JSP
- Spring
- Spring Boot
- Struts
- JSF

Non-Java technologies

- PHP
- Python
- Node.js
- Ruby

For your Java backend journey, the important path is:

Java
↓
Servlet
↓
Spring
↓
Spring Boot

Important Points to Remember

Servlet

→ Java program that runs on a server.

Client

→ Sends requests.

Server

→ Receives and processes requests.

Tomcat

→ Server used to run Java web applications and Servlets.

Servlet Container

→ Manages and runs Servlets.

Request

→ Client asks the server to do something.

Response

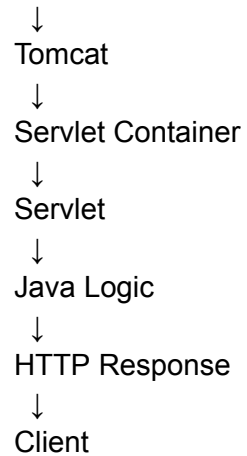
→ Server sends the result back to the client.

The most important flow

Client

↓

HTTP Request



In one sentence:

A Servlet is a Java program that runs inside a Servlet Container such as Tomcat, receives client requests, processes them, and sends responses back to the client.

Lecture link:

https://www.youtube.com/watch?v=WkKT5M-ABnY&list=PLIhM4Ikb2sEjVsbbZ_kiixY5CcR84IQUg&index=5